

AI Intelligence Pipeline — System Architecture

Production-oriented architecture for large-scale AI venture intelligence ingestion

Sources YC AI · Best of AI · AI/FOXX · arXiv · RSS	Async Crawler aiohttp · bounded concurrency · retry	Extraction HTML/RSS → clean text → fields
LLM Orchestration Gemini → Groq → DeepSeek	Entity Resolution domain/name/evidence scoring	Quality & Freshness 24h filter · dedupe · schema checks
Storage raw + processed JSON/CSV	Export 6 Google Sheet datasets	Observability failures · checkpoints · mapping log

1. Architecture and Data Flow

The pipeline separates acquisition, extraction, LLM normalization, entity resolution, freshness validation and export. Raw source responses are kept separately from processed records so extraction logic can be rerun without repeatedly downloading sources. Every accepted record retains a legitimate source URL for traceability.

- Source discovery creates URL queues for startups, products, papers, jobs and news.
- Asynchronous workers crawl URLs concurrently with bounded concurrency, timeouts and retries.
- Extractors remove scripts/navigation and convert pages or feeds into compact text.
- Canonical JSON is validated before persistence; rejected and failed URLs are tracked.

2. Scaling to 500,000+ Records

The design is queue-oriented and horizontally scalable. URL discovery feeds partitionable work queues; workers process independent batches; records are persisted incrementally. No stage relies on a fixed dataset size. Scale is achieved by adding workers and increasing partitions while preserving per-host limits.

Layer	Scale strategy
Discovery	Stream/paginate source enumeration; deduplicate URL queue.
Crawling	Multiple async workers, connection pooling, per-host limits and retry queues.
LLM	Provider fallback, safe truncation/chunking, retries and caching.
Processing	Batch checkpoints, idempotent keys and separate raw/processed stores.
Storage	Partition by source/date/entity; indexes on URL, normalized name and content hash.

3. Robustness: 413, 429 and Source Protection

- 413 Payload Too Large:** reduce input before submission; for production, split long documents into semantic chunks, extract each chunk, then merge and validate results.
- 429 Rate Limit:** use exponential backoff with jitter for transient limits. Provider health is tracked; exhausted providers are skipped and the next provider is attempted.
- Cloudflare/Datadome/JS protection:** no illegal bypass is attempted. Blocked pages are recorded; permitted RSS/public structured endpoints are used when available and processing continues.

4. Distributed Freshness, Idempotency and Deduplication

- Each URL receives a deterministic work key such as normalized URL + source. A queue lease prevents concurrent duplicate processing. Content hashes and canonical URL indexes provide a second duplicate barrier. Publication timestamps are normalized and news/jobs are filtered to the required 24-hour window. Processing status supports retries and crash recovery.
- Recommended production storage: Redis/Kafka/SQS for work queues; PostgreSQL for canonical records; object storage for raw HTML; pgvector/vector DB for semantic retrieval; and Neo4j or another graph layer for startup–product–paper relationships.

5. LLM Orchestration and Canonicalization

Extraction uses a strict JSON contract. Models are instructed to use only information explicitly present in source text, return null when unavailable, reject unrelated pages, and emit JSON only. Responses are parsed and schema-validated before acceptance.

Priority	Provider / role	Failure handling
1	Gemini Flash — primary	Disable after permanent/quota failure; continue.
2	Groq Llama-compatible endpoint — fallback	Retry temporary rate limits; switch provider when unavailable.
3	DeepSeek — secondary fallback	OpenAI-compatible client; skipped when unavailable.

6. Deterministic Entity Resolution

Product-to-startup mapping uses auditable evidence scoring. Exact domains receive the strongest evidence, followed by registered domains, known company names and description/company evidence. Ambiguous or low-confidence cases remain unresolved instead of being guessed. The Entity Mapping Log records status, confidence and match method for every product.

Current resolution: 1,672 products → 466 matched (27.87%) and 1,206 unresolved. Unresolved records are intentionally retained for future enrichment.

7. Current Validated Dataset

Dataset	Records	Validation result
Startups	1,028	1,028 unique names; 0 duplicate URLs; 1,024 official websites
Products	1,672	Valid pricing models; 466 mapped; 1,206 unresolved
Research Papers	1,000	Paper URL, GitHub URL and GitHub stars present
Jobs	43	43 fresh within 24h; required fields present
News	4	4 fresh within 24h; no stale/missing dates
Entity Mapping Log	1,672	466 matched + 1,206 unresolved; 0 invalid statuses

8. Quality Gates and Reproducibility

The final validator checks counts, uniqueness, required fields, URL validity, freshness, pricing enums, GitHub metrics, entity-resolution consistency and mapping-log integrity. The completed run passed all 7 validation groups. Raw/processed files, failures and mapping decisions are retained for audit and reruns.

OVERALL VALIDATION STATUS: PASS — 7/7 CHECKS

9. Production Evolution

For higher scale, deploy crawler workers as containers, move URL work to a durable queue, store raw documents in object storage, use PostgreSQL/warehouse tables for canonical data, and add vector/graph indexes asynchronously. Add metrics for queue depth, crawl latency, HTTP status distribution, provider errors, extraction acceptance and duplicate rate. These improvements preserve the existing data contracts.